Image Analysis

An Open-Source Summary

Thomas Debelle

KU LEUVEN

April 20, 2026

Contents

1 Introduction

The key take away is that vision is subjective to the human being and themselves. It is the results of continuous evolution and survival of the fittest. We are more sensitive to light and certain colors, to motion, ... And this fact will be used for efficient image representation and analysis.

We can think of the classic illusion where our brain will interpret certain lengths of an arrow longer than others. Context also does matter, even in a blurry picture our brain will train to make up for what he sees.

1.1 The basics

We first need light to have vision. Light is influenced by object and create perception. ADD image slide 51. The light source bounce on the object and some scattered ray will get trough the 2D Image plane, which will be casted on the sensor array thanks to the lens system.

1.1.1 Light

We will look at it from a physical optic (not geometrical (constant speed and straight path) nor quantum). Light is an Electro-magnetic wave which we assume it to be planar. aka $\frac{|E|}{|H|} = \eta$ in simple term, light is perpendicular. This is a **self-sustaining process** that is characterized by:

1. λ: wavelength
2. Direction
3. E: Amplitude
4. φ: phase
5. Direction of polarisation

Violet: low frequency ($380nm$); Red: high frequency ($760nm$). Remember $\lambda = \frac{c}{f}$.

1.1.2 Perception

We don't perceive light equally. Humans have 3 cones that are not evenly spread apart. But the human vision is trained to compensate for it so everything feels similar in intensity to us. Some animals can have more cones.

1.1.3 Interaction with matter

Table 1: The four types of interaction

Phenomenon	**Example**
Absorption	Blue water,green leaf
Scattering	blue sky, red sunset
Reflection	colored in k
Refraction	dispersion by a prism

1.1.3.1 Absorption Dissipation of λ specific for the medium. Resonance in molecules.

1.1.3.2 Scattering When light (radiation/particles) deviates from a straight trajectory by local non-uniformities. Type depends on the relative size of particles and wavelengths:

1. **Rayleigh**: small (λ dependent)
2. **Mie**: comparable (weakly λ dependent)
3. **Non-selective**: large (λ **in**dependent)

We ignored transmission and diffraction.

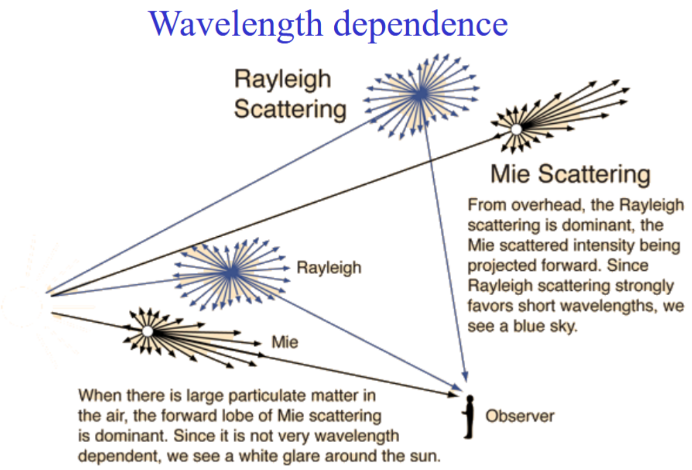

Figure 1: Wavelength dependence for scattering

We need scatter to get the bluesky (Rayleigh with Tyndall effect) or it would be dark. Coloured cloud from volcanic eruption = Mie (particle all over the air). Grey cloud is due to non-selective.Less scatter for lower frequency light (infrared).

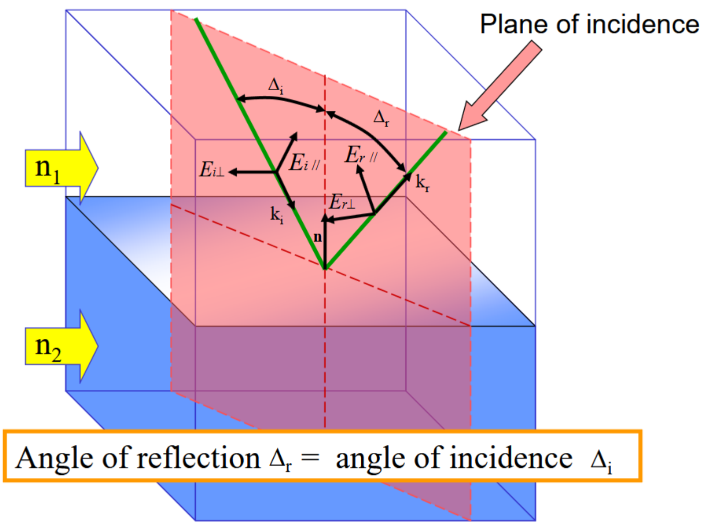

Figure 2: Mirror reflection

1.1.3.3 Reflection With reflection, the notion of polarization is important. The polarization effect creates the fact that a reflected wave on certain surfaces can be reflected with all but one direction of the electric field. This effect is more prominent on **non-metallic** surfaces!

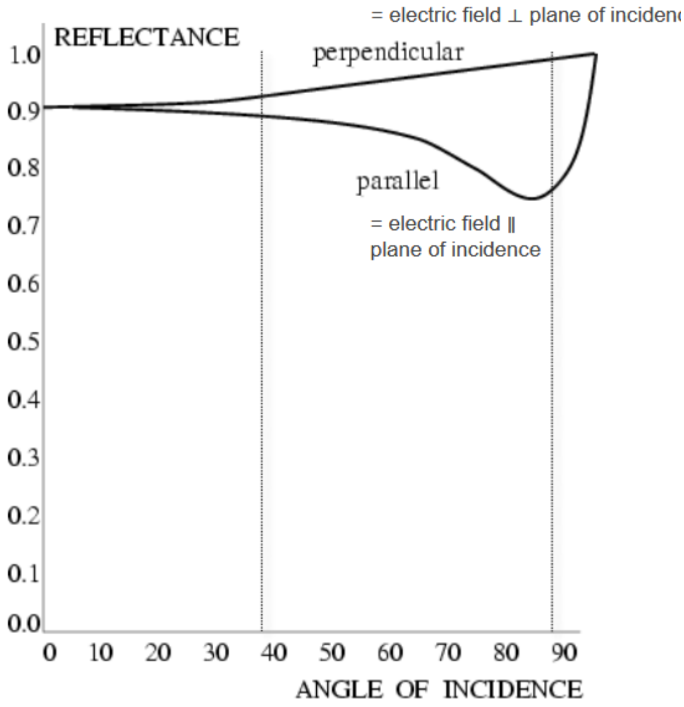

Figure 3: Non-metallic reflection - strong reflectors

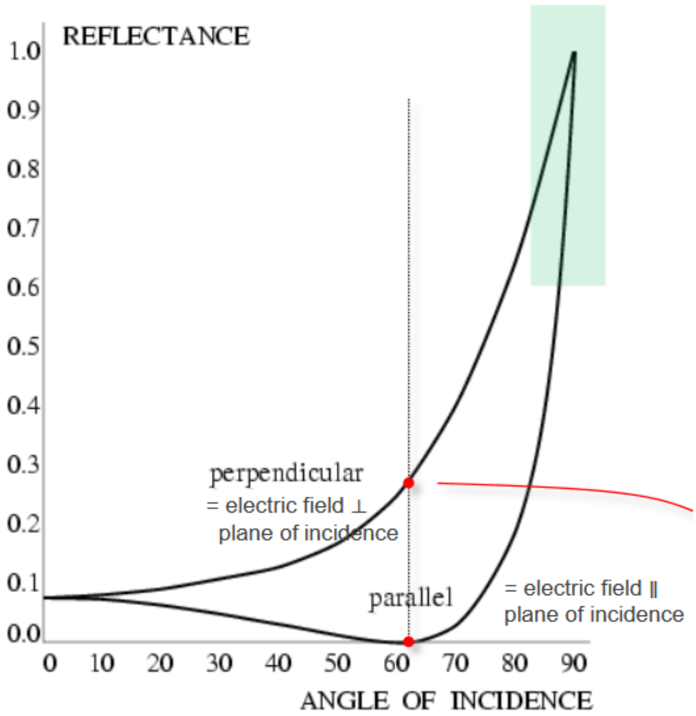

Figure 4: Metallic reflection - Red dot: Brewster angle (polarizer) - Blue zone: Grazing angles

Rough surfaces will lead to **diffuse** reflection as the reflection angles will be all over the place. On the other hand, smooth surfaces give **specular** reflection. Both **diffuse** and **specular** can also happen all together.

Reflectance varies through λ and different side of vegetation can typically give various reflection[1]

1.1.3.4 Refraction Typically, refraction which is the bending of the light and dispersion which is a frequency dependency.

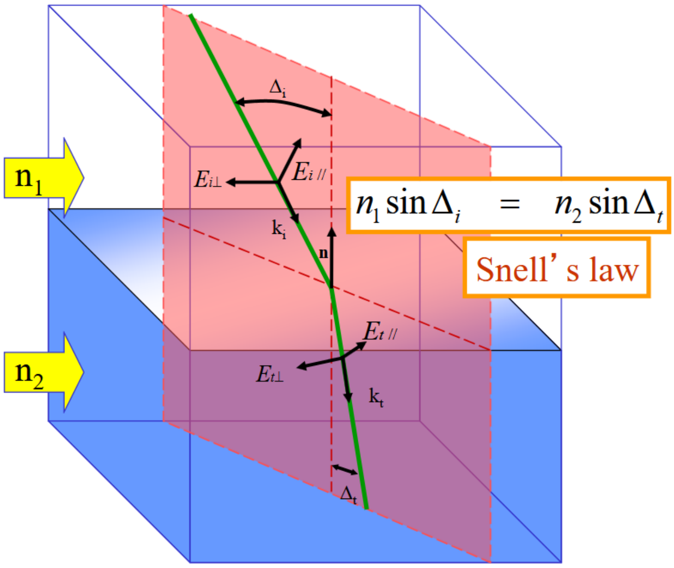

Figure 5: Refraction - governed by Snell's law property of the medium **and** material

2 Acquisition of images

Controlling the lights is essential for acquisition as it is the main vector to acquire images. Severals techniques exist

Type	Description	Examples
Back-lighting	Light source behind object (better with a diffuser plate)	High contrast silhouette. **Binary vision**, inspection
Directional	Tilt lights to generate sharp shadow of specular reflection (rough surfaces)	Detection of cracks
Diffuse	Uniform lighting, all directions contribute to the diffusion reflection but specular spike due to spike	Detection of cracks
Polarized: Contrast diffuse and specular	diffuse makes light non-polarized while specular keeps polarized. Put polarizer/analyzer orthogonal, better for glare (high dynamic range) issue.	Detection of cracks

[1] This fact can be used for spectral reflectance of vegetation since it forms a signature

Type	Description	Examples
Polarized: Contrast dielectrics and metal	Dielectric at Brewster angle can be used as a polarizer which is not possible for metallic surface (see previous chapter).	So use a polarizer to suppress specular reflection from dielectric **OR** distinguish between specular and dielectric surfaces
Coloured		
Structured		
Stroboscopic		

Note about the dar and bright field. Mostly relevant for the Directional/Diffuse lighting. *Dark field*, when the camera is placed outside the specular reflection area of the normal area, only specular reflection (different orientation from normal) will show up. On the other hand, *Bright field*, we place camera inside this specular reflection light area and the non-normal area will appear dark.

3 License

3.1 Modification

To contribute to this work or any other one from this project please find more information at the Github repository.

3.2 Take down

If you are a professor or representing any official party and wish to take down this work, you can submit a takedown request at this url: https://github.com/Tfloow/ESATSummary/issues/new?template=takedown_notice.yml